

University of Essex

MSc Artificial Intelligence

Module: IA_PCOM7E January 2026

Unit 11: Collaborative Discussion – Ethics of Deep Learning

Initial Discussion

The topic under discussion focuses on the potential ethical issues surrounding the use of deep learning models to generate text and images.

In my view, and in the view of much of the current state of the literature, there are numerous ethical pitfalls to consider, including:

1. The risk of a model propagating known false information is non-zero, with subsequent ethical considerations arising from the consequences of generating and propagating that same false information (Kumar and Poonam, 2025; Sivagnanam et al, 2025).
2. Innate societal biases can be inadvertently incorporated into the model, skewing its outcomes and practical utility (Kumar and Poonam, 2025; Sivagnanam et al, 2025).
3. In the legal context, appropriate permission to use, and appropriate attribution of, source training material may be compromised by negligent model development or by model development undertaken by unethical actors intentionally disregarding creator intellectual property ownership rights (Kumar and Poonam, 2025; Sivagnanam et al, 2025).
4. End users who misunderstand how a model works, or the limits of its capabilities, may rely on generated material to their psychological, financial, or other real world detriment (Kumar and Poonam, 2025; Sivagnanam et al, 2025).

In other words, generative models carry the same set of ethical challenges as any other source of intellectual property content, albeit in what many people unfamiliar with the technology may consider to be a new format or a novel mode of dissemination.

References

Kumar, K. and Poonam, N. (2025). Responsible AI: Ethical Considerations in Generative AI. *Textual Intelligence: Large Language Models and Their Real-World Applications*, pp. 413-434. doi:<https://doi.org/10.1002/9781394287499.ch17>.

Sivagnanam, B., Rajiakodi, S., Pannerselvam, K. and Chakravarthi, B. (2025). Ethical Dimensions and Societal Effects of Generative AI: The Portrayal of Ethical Issues of ChatGPT, DALL-E, and Other Systems. *Generative AI: Disruptive Technologies for Innovative Applications*, pp. 131-151. doi:<https://doi.org/10.1002/9781394302932.ch7>.

Peer Response No. 1

Your post effectively summarises the top ethical challenges that have risen with the advancement of generative Artificial Intelligence (GenAI). Misinformation, amplifying societal biases, copyrights infringement and inappropriate use are all areas to question how we can reap the benefits of GenAI while navigating the risks of it (Kumar and Poonam, 2025; Sivagnanam et al, 2025).

The risk that is most relevant to the one that I chose is the one related to copyrights, or rather the risks related to GenAI's ability to 'create'. Since GenAI can create new content in all forms (Sternberg, 2024), a question rises regarding what human creativity means in this new era. After computers took over the advantage on the front of computational capabilities since the 1970s, creativity has long been considered a capability very specific to human beings. Now, this belief is threatened with founded suspicions.

However, adding your concern to mine, the issue becomes double folded. GenAI is shaking the claim of uniqueness of human creativity, while it is also generating content that is based on humans' content in the first place, of which copyrights might be violated by GenAI content.

I think the answer to your concern and mine – and to them both combined – can be mainly either of two approaches. One is to disregard AI-generated content and continue to wage wars to prohibit the training of generative deep learning models on human artefacts, with all the doubts of what that will lead to. The second is to shift our focus to build more robust and sustainable human-machine collaboration frameworks, which would require a revisit to all of our standing concepts, including copyrights, AI companies' legal guardrails, and what creativity means (Cai and Gao, 2025).

References

Cai, W. and Gao, M. (2025) Beyond Hallucination: Generative AI as a Catalyst for Human Creativity and Cognitive Evolution, *ICCK Transactions on Emerging Topics in Artificial Intelligence*, 2(1), pp. 36–42. Available at: <https://doi.org/10.62762/tetai.2025.657559>.

Kumar, K. and Poonam, N. (2025). Responsible AI: Ethical Considerations in Generative AI. *Textual Intelligence: Large Language Models and Their Real-World Applications*, pp. 413-434. doi:<https://doi.org/10.1002/9781394287499.ch17>.

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Technologies for Innovative Applications, pp. 131-151. doi:<https://doi.org/10.1002/9781394302932.ch7>.

Peer Response No. 2

You've provided a clear and well-structured overview of the key ethical issues surrounding generative AI, particularly in highlighting misinformation, bias, and intellectual property concerns. I agree with your point that these challenges are not entirely new but are intensified by the scale and speed at which deep learning models can generate and distribute content.

One aspect I found particularly important in your post is the role of user misunderstanding. As you noted, many users may overestimate the reliability of AI-generated outputs, which can lead to real-world consequences. Building on this, I think there is a strong need for better transparency in how these systems communicate their limitations, such as clearly indicating uncertainty or potential inaccuracies in generated content.

Additionally, while you mention that generative AI shares similar ethical challenges with traditional content, I would argue that its ability to automate and personalise content at scale makes these issues more complex and harder to regulate. For example, misinformation can now be produced and targeted much more efficiently than before.

References

No references were provided with this peer response.

Discussion Summary

The topic under discussion focused on ethical issues arising from the use of deep learning models to produce text and images (for brevity, “content”).

I asserted several potential areas of ethical concern:

1. The risk of content containing, or have the effect of promoting, false information (whether intentionally or inadvertently) is significant (Kumar and Poonam, 2025; Sivagnanam et al, 2025).
2. Likewise, the risk of unmediated ideological biases of a variety types being propagated uncritically is also significant. (Kumar and Poonam, 2025; Sivagnanam et al, 2025).
3. Legal pitfalls surrounding intellectual property ownership rights of source material and the content derived from it are numerous (Cai and Gao, 2025; Kumar and Poonam, 2025; Sivagnanam et al, 2025).
4. End users who don't know, or who misunderstand, how the content was created and how best to interpret it in an appropriately objective, contextually oriented manner can be misled to their detriment (Kumar and Poonam, 2025; Sivagnanam et al, 2025).

Peer responses were in alignment with these assertions and focused on the second and third assertions.

For the second assertion, a peer response asserted that overreliance by end users on the assumed accuracy of content can be mitigated by requiring greater transparency from content creators about the source material and developmental mechanisms from which they generated it.

For the third assertion, a peer response noted that systems for intellectual property rights are still in flux on the subject of how much protection, if any, may be granted to such content. More broadly, the peer response also raised the philosophical concern that the inherent nature of such artificially generated content calls for renewed inquiry into the nature of human creativity, and which aspects of established legal protections for human creations may appropriately transfer over to artificial creations.

I agree with the substance of both peer responses: greater transparency and further inquiry into these subject matters should be useful avenues of exploration for framing solutions to the ethical concerns they raise.

References

Cai, W. and Gao, M. (2025) Beyond Hallucination: Generative AI as a Catalyst for Human Creativity and Cognitive Evolution, *ICCK Transactions on Emerging Topics in Artificial Intelligence*, 2(1), pp. 36–42. Available at: <https://doi.org/10.62762/tetai.2025.657559>.

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